

The polluted atmosphere as a shallow domain

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The shallow domain concept

- Natural, large-scale geophysical fluids share the shallow domain characteristics
- "almost" two-dimensional set $\rightarrow$ we can significantly simplify the model in the vertical direction
- a domain that is excessively greater horizontally than in the vertical dimension $\rightarrow$ leading to the classic hydrostatic approximation of the Navier-Stokes equations

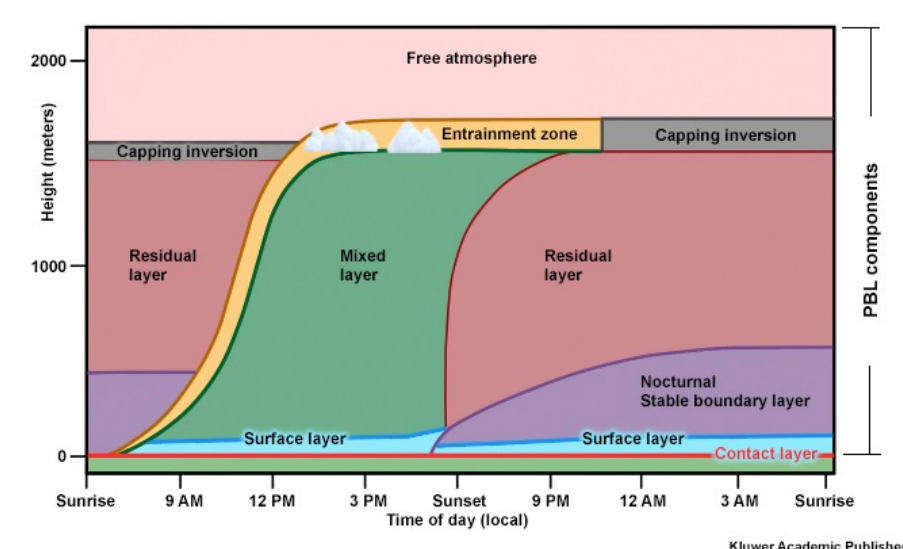


Fig. 1: A local slice of the atmosphere. Picture credit: mountwashington.org

The main ideas and tools in this scenario:

- introduce the aspect ratio $\epsilon = \frac{\text{characteristic depth}}{\text{characteristic width}}$
- anisotropic viscosity hypothesis,
- rescaling, $x = x_1, y = x_2, z = \epsilon x_3$.

Correspondingly, we rescale

- the kinematic terms: $v_3 = \epsilon u_3^\epsilon$,
- the viscosity terms: $v_z = \epsilon^2 v_3$,
- the diffusivity terms: $K_x = \epsilon K_1, K_y = K_2, K_z = \epsilon^2 K_3$.

Modelling - challenges and choices

- Coordinate system $\rightarrow$ we fix it in a way so that x and y match the downwind and crosswind direction, respectively (while z represents the vertical dimension).
Natural assumption in the downwind direction: diffusion is negligible compared to downwind transport, i.e.

$$|u_1 \partial_{x_1} C| \gg \left| K_x \frac{\partial^2 C}{\partial x_1^2} \right|,$$

so we can take advantage of the additional scaling equation $K_x = \epsilon K_1$.

- Domain $\rightarrow$ we use locally acceptable Cartesian coordinates — this allows us to ignore large-scale effects such as the curvature of the Earth.

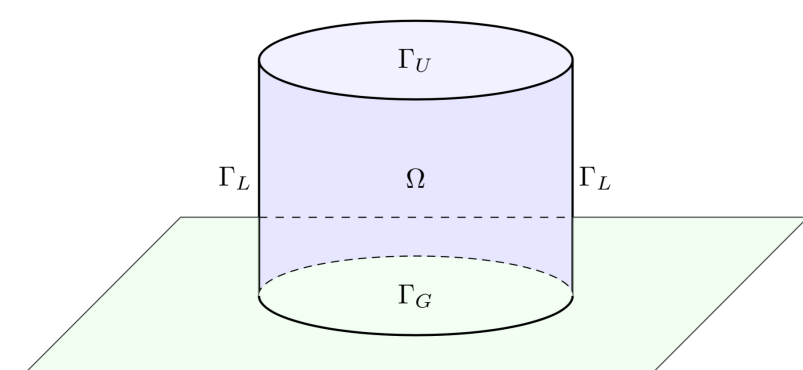


Fig. 2: A local slice of the atmosphere

Challenges: local area model $\rightarrow$ open boundary conditions on the lateral boundary. We use a fix zero approach — the concentration being set to zero actually stands for a constant concentration with the background flow being removed.

- Source function of the concentration $\rightarrow$ Gaussian impulse approach. We use a parametrized nascent delta function to describe the source term, with the parameter being ϵ . We have $S_\epsilon(t, t_s, x, x_s) = s(t, t_s) \delta_\epsilon(x - x_s)$, with the activation term $s(t, t_s) = IH(t - t_s)$ where $IH(\cdot)$ is the Heaviside step function. In the limit model we naturally arrive to the delta distribution as a source.

The anisotropic equations

$$\partial_t u_1^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_1^\epsilon - \Delta_\nu u_1^\epsilon - \alpha u_2^\epsilon + \epsilon \beta u_3^\epsilon + \partial_1 p^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t u_2^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_2^\epsilon - \Delta_\nu u_2^\epsilon + \alpha u_1^\epsilon + \partial_2 p^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\epsilon^2 \{ \partial_t u_3^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon - \Delta_\nu u_3^\epsilon \} - \epsilon \beta u_1^\epsilon + \partial_3 p^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\nabla \cdot \mathbf{u}^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t C^\epsilon + \mathbf{u}^\epsilon \cdot \nabla C^\epsilon = \epsilon K_1 \frac{\partial^2 C^\epsilon}{\partial x_1^2} + K_2 \frac{\partial^2 C^\epsilon}{\partial x_2^2} + K_3 \frac{\partial^2 C^\epsilon}{\partial x_3^2} + S_\epsilon \text{ in } \Omega \times (0, T)$$

$$\mathbf{u}^\epsilon = 0 \text{ on } \Gamma \times (0, T)$$

$$\mathbf{u}^\epsilon(\cdot, t = 0) = \mathbf{u}_0, C^\epsilon(\cdot, t = 0) = C_0^\epsilon \text{ in } \Omega$$

$$C^\epsilon = 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0, T), \partial_2 C^\epsilon = \partial_3 C^\epsilon = 0 \text{ on } \Gamma_G \times (0, T), \partial_1 C^\epsilon \in L^2 L^2(0, T; \Gamma_G)$$

The original equations We use the incompressible Navier-Stokes equations in all three space dimensions for the fluid velocity.

The limit model

In the vertical dimension we arrive to the hydrostatic approximation.

$$\partial_t u_1 + \mathbf{u} \cdot \nabla u_1 - \Delta_\nu u_1 - \alpha u_2 + \partial_1 p = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t u_2 + \mathbf{u} \cdot \nabla u_2 - \Delta_\nu u_2 - \alpha u_1 + \partial_2 p = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_3 p = 0 \text{ in } \Omega \times (0, T)$$

$$\nabla \cdot \mathbf{u} = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t C + \mathbf{u} \cdot \nabla C = K_2 \frac{\partial^2 C}{\partial x_2^2} + K_3 \frac{\partial^2 C}{\partial x_3^2} + S \text{ in } \Omega \times (0, T)$$

$$u_1 = u_2 = u_3 n_3 = 0 \text{ on } \Gamma \times (0, T)$$

$$u_i(\cdot, t = 0) = u_{0i}, C(\cdot, t = 0) = C_0 \text{ in } \Omega, i = 1, 2$$

$$C = 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0, T), \partial_2 C = \partial_3 C = 0 \text{ on } \Gamma_G \times (0, T), \partial_1 C \in L^2 L^2(0, T; \Gamma_G)$$

The main theorem

Theorem

Let $\mathbf{u}_0 \in L^2(\Omega)^3$, with $\nabla \cdot \mathbf{u}_0 = 0, \mathbf{u}_0 \cdot \mathbf{n} = 0$ on $\partial\Omega$; and let $C(0) \in L^2(\Omega)^3$, $C(0) = 0$ on $\partial\Omega$; there exists a weak solution $(\mathbf{u}, C)$ of the hydrostatic equations of the polluted atmosphere, obtained as a limit of weak solutions $(\mathbf{u}^\epsilon, C^\epsilon)$ of the anisotropic equations as the aspect ratio ϵ tends to 0.

A priori estimates, weak convergence

The main steps for proving the weak convergence are the following.

- Energy inequality.

$$\begin{aligned} & \frac{1}{2} \{ \|u_{1t}^\epsilon(t)\|_{L^2}^2 + \epsilon^2 \|u_{3t}^\epsilon(t)\|_{L^2}^2 + \|C^\epsilon(t)\|_{L^2}^2 \} + \int_0^t \{ \|\nabla_\nu u_{1t}^\epsilon\|_{L^2}^2 + \epsilon^2 \|\nabla_\nu u_{3t}^\epsilon\|_{L^2}^2 \} d\tau + \\ & \int_0^t \{ (\epsilon K_1 - \nu_2^\epsilon K_{1CT}(c_S + 1)) \|\partial_1 C^\epsilon\|_{L^2}^2 + (K_2 - \nu_2^\epsilon K_{1CT}(c_S + 1)) \|\partial_2 C^\epsilon\|_{L^2}^2 + \\ & + (K_3 - \nu_2^\epsilon K_{1CT}(c_S + 1)) \|\partial_3 C^\epsilon\|_{L^2}^2 \} d\tau \leq \frac{1}{\gamma} K_{12} \int_0^t \|\partial_1 C^\epsilon\|_{L^2(\Gamma_G)}^2 + \\ & + \frac{1}{2} \{ \|u_{1t}^\epsilon(0)\|_{L^2}^2 + \epsilon^2 \|u_{3t}^\epsilon(0)\|_{L^2}^2 + \|C^\epsilon(0)\|_{L^2}^2 \} + \int_0^t (S^\epsilon, C^\epsilon) d\tau. \end{aligned}$$

- A priori estimates: $L_t^\infty L_x^2$ boundedness for C^ϵ ; $L_t^2 L_x^2$ boundedness for $\sqrt{\epsilon} \partial_1 C^\epsilon, \partial_2 C^\epsilon, \partial_3 C^\epsilon$.
- Weak convergence of the linear terms.
- Weak convergence of nonlinear terms.
 - $\rightarrow$ Nonlinear terms not including the C^ϵ term - divergence free condition
 - $\rightarrow$ The term $(\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C})$.

We need a compactness result to obtain $u_i^\epsilon C^\epsilon \rightharpoonup u_i C$. For this we show that

- we have a bound for the time derivative of the concentration:

$$\frac{\partial C^\epsilon}{\partial t} = -\mathbf{u}^\epsilon \cdot \nabla C^\epsilon + \sqrt{\epsilon} K_1 \partial_1 \partial_1 \sqrt{\epsilon} C^\epsilon + K_2 \partial_2 \partial_2 C^\epsilon + K_3 \partial_3 \partial_3 C^\epsilon - S^\epsilon,$$

Eventually we have $\frac{\partial C}{\partial t} \in L^1 W^{-1,1}$,

- we have an equicontinuity for velocity: $\|u_i^\epsilon - u_i^\epsilon(\cdot + \xi, t)\|_{L^2 L^2} \rightarrow 0$ as $\xi \rightarrow 0$.

Combining these results allow us to apply a compactness theorem that provides the convergence.

Additional results

- We also provide an existence theorem for the extended system.
- We have generalized the main theorem for the seaside case, when there is interaction between water and air $\rightarrow$ different boundary conditions.